The Illusion of Accuracy: Auditing CNN Shortcut Learning with Grad-CAM & Occlusion

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GITHUB LINK: https://github.com/rohitha-12/MachineLearning.git

1. Introduction: The "Clever Hans" Dilemma in Artificial Intelligence

Deep learning models are extensively utilized across various industries, frequently achieving near-perfect accuracy on validation datasets. Nonetheless, regarding a Convolutional Neural Network (CNN) as an infallible "black box" constitutes a perilous approach. Elevated accuracy does not ensure that a model has genuinely comprehended the fundamental target concept; it frequently merely demonstrates that the model has identified a statistical correlation to the proper label. This route is often a deceptive "shortcut."

In the machine learning community, this phenomenon is referred to as the "Clever Hans" effect. In the early 20th century, a horse named Clever Hans gained international fame for ostensibly executing intricate arithmetic calculations. In truth, Hans was not engaged in mathematics; he had only acquired the ability to interpret the subtle, unconscious alterations in his trainer's body language to discern when to cease tapping his hoof.

Neural networks exemplify the contemporary iteration of Clever Hans. A CNN may accurately classify a "speedboat" with 99% confidence, not because to its recognition of the hull's geometric shape or the engine, but rather because it has only memorized the blue and green pixels of the water in the training data's background. Presenting the identical model with a boat situated on dry land, or an entirely different geometric figure against a green backdrop, will result in a catastrophic failure.

To develop resilient and reliable AI, we must not depend solely on accuracy measures. We must conduct thorough audits of our networks. This lesson will employ Gradient-weighted Class Activation Mapping (Grad-CAM) and Occlusion Sensitivity Testing to elucidate the inner workings of our CNN, show its focus, and reveal its concealed contextual biases.

2. The Theory: How Grad-CAM Preserves Spatial Reasoning

Conventional saliency maps employ fundamental backpropagation to emphasize pixel-level significance; however, these results frequently exhibit excessive noise and granularity, rendering them difficult to comprehend. Grad-CAM (Gradient-weighted

Class Activation Mapping) addresses this issue by interfacing with the final convolutional layer of the network instead of analyzing individual input pixels.

What is the purpose of the final convolutional layer? Convolutional layers preserve spatial information. When data enters the fully linked (dense) layers at the conclusion of a CNN, its spatial configuration is transformed into a 1D vector and irretrievably lost. Consequently, the ultimate convolutional feature maps embody the most intricate, advanced spatial ideas (such as "eyes" or "wheels") that the model has assimilated prior to rendering a conclusion.

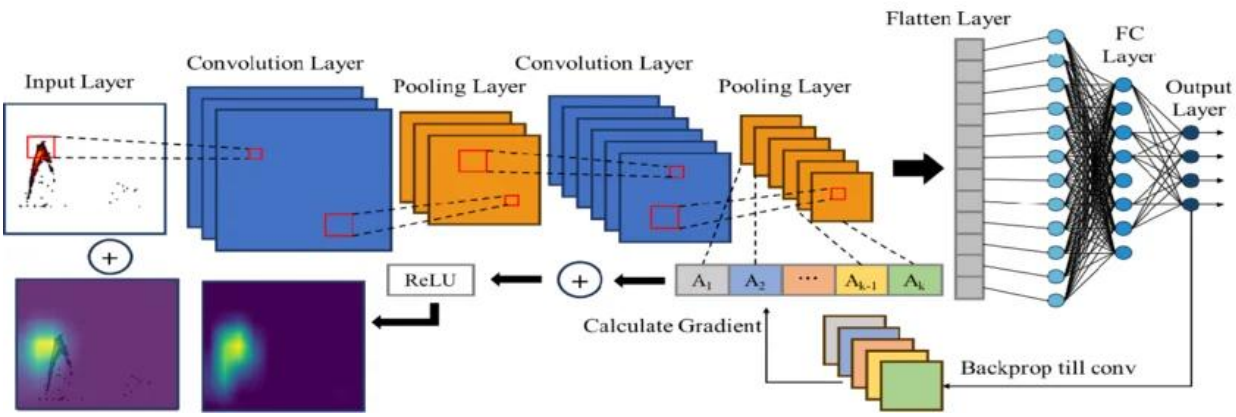

Figure: The Grad-CAM Architecture Flow
A flowchart depicting the Grad-CAM methodology. An input image traverses convolutional layers to generate feature maps. The gradient of the target class score is calculated about these feature maps. The gradients are subjected to global average pooling to generate weights, which are then multiplied by the forward activation maps. A ReLU function is ultimately employed to produce the coarse localization heatmap.

Grad-CAM mathematically calculates the gradient of the target class score (y^c) in relation to the feature mappings (A^k) of the final convolutional layer. The gradients are subjected to global average pooling to derive importance weights (α_k^c):

$$\alpha_k^c = \frac{1}{Z}\sum_i \sum_j \frac{\partial y^c}{\partial A_{i,j}^k}$$

Consider α_k^c as the "vote" of a certain feature map. If a feature map identifies "fur," and the target class is "Dog," the feature map receives a substantial positive weight.

The weights are subsequently multiplied by the feature maps, followed by the application of a ReLU activation to retain just the features that positively impact class prediction:

$$L^c_{Grad-CAM} = ReLU\left(\sum_k \alpha^c_k A^k\right)$$

Utilizing the ReLU function, we exclude any pixels associated with other classes, resulting in a refined, localized heatmap that precisely illustrates the basis of the network's decision-making process.

3. Methodology & Strict Accessibility Standards

This lesson utilizes a dual approach to comprehensively evaluate the decision-making process of our deep learning model. We will evaluate a pre-trained ResNet50 Convolutional Neural Network (CNN) with conventional ImageNet weights. Our aim is to examine the network's behavior under standard conditions and, importantly, its response when its anticipated visual elements are forcibly eliminated.

Initially, we create our baseline with a bespoke Python implementation of Grad-CAM. We employ PyTorch hooks to capture the forward activation maps and backward gradients from the network's last convolutional layer (layer4[-1].conv3). This enables us to visualize the model's untainted spatial focus.

Secondly, we increase the technical complexity by implementing an Occlusion Sensitivity Test. Rather of passively viewing the model, we employ automated scripts to superimpose large, opaque black bounding boxes over the principal subjects in our test photographs. By reintroducing these purposefully compromised images into the Grad-CAM pipeline, we compel the model to disclose its genuine rationale: does it diligently seek out residual fragments of the object, or does it resort to inappropriate background heuristics?

Accessibility as a Fundamental Research Criterion

Interpretability methods are inherently visual, hence historically excluding visually handicapped researchers or readers. A heatmap is ineffective if the audience cannot precisely identify the activation zones. To guarantee the universal accessibility of this lesson, we have established rigorous visual formatting requirements in our code:

Perceptually Uniform Colormaps: The utilization of the default 'Jet' colormap is explicitly forbidden, as it generates false visual demarcations and is illegible for individuals with red-green color vision deficits (CVD). All heatmaps in this tutorial employ the viridis colormap. This palette uniformly enhances luminance, guaranteeing the heatmap's readability across all types of colorblindness and in grayscale printing.

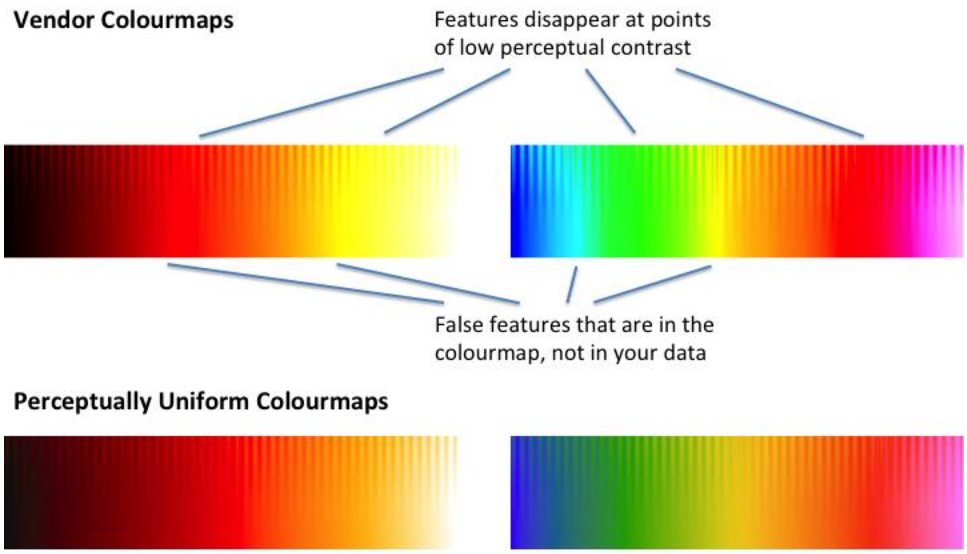

Figure: Accessibility Failure in Standard Heatmaps
A visual juxtaposition of the 'Jet' and 'Viridis' colormaps. The graphic illustrates how the 'Jet' colormap produces unnatural, discordant streaks of color that become indistinct and illegible when subjected to a red-green colorblindness simulation. Conversely, the 'Viridis' colormap preserves a seamless, discernible gradation from dark to bright across all viewing situations.

Dual-Coding (Geometric Isolation): Information should not be communicated solely through color. Our own graphing algorithm systematically identifies the top 20% of network activations and delineates this central focal area with a prominent, high-contrast white bounding box.

This dual-coding delineates the precise area of the network's focus by a distinct geometric shape and a color gradient, meeting the utmost criteria for academic accessibility.

4. Experiment 1: Establishing the Baseline Feature Audit

Prior to identifying the deficiencies of a deep learning model, it is essential to define what constitutes a valid, mathematically robust prediction. We require a "control" for our experiment. This baseline audit assesses the performance of our pre-trained ResNet50 model on a standard, unmodified image in which the subject is distinctly isolated from its surrounding environment.

For this evaluation, we provided an image of a Samoyed dog (ImageNet Class 258) positioned on a green lawn. The model accurately predicts the appropriate class. Nevertheless, as demonstrated by the Clever Hans phenomenon, we cannot regard this accuracy statistic uncritically. We must ascertain the rationale behind this decision. Did it

genuinely identify the anatomical characteristics of the dog, or did it only acquire a misleading training association between white pixel clusters and green grass?

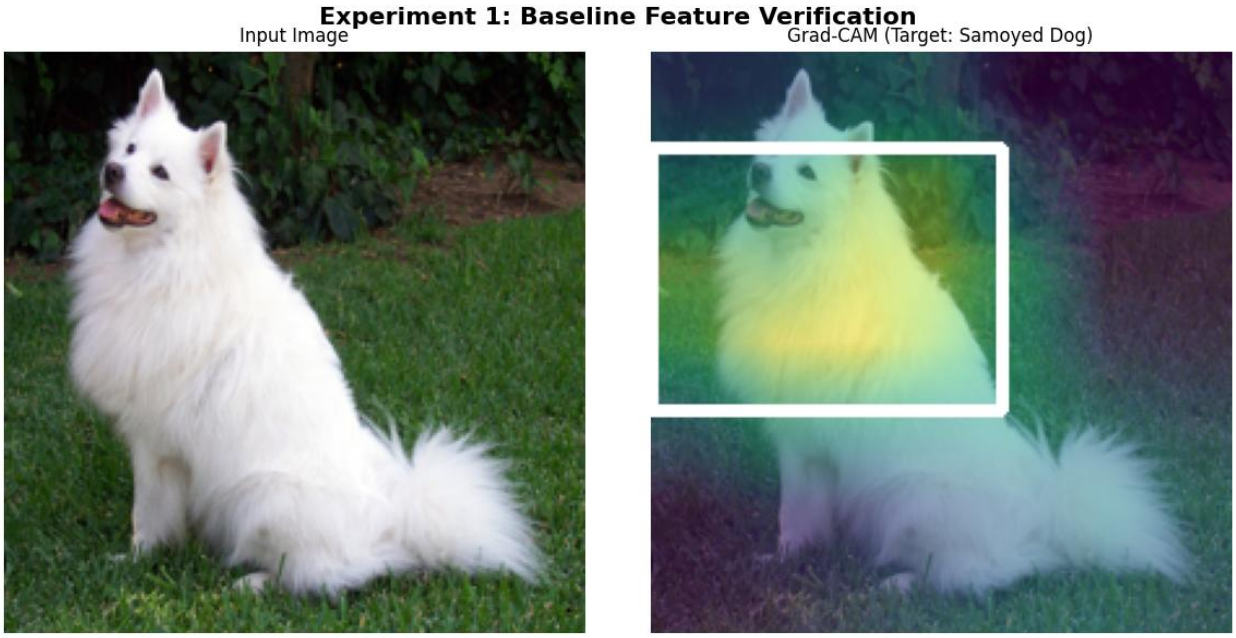

Figure 1 Analysis: The Healthy Baseline
A comparative diagnostic plot. The left panel depicts a white Samoyed dog seated on verdant grass. The right panel exhibits the Grad-CAM overlay with a perceptually uniform Viridis colormap. A robust white bounding box dynamically surrounds the most luminous yellow activations, which are predominantly located on the dog's chest, neck, and lower nose.

The visualization demonstrates that the model operates accurately and securely. The Grad-CAM heatmap indicates that the network's spatial attention is only influenced by the fundamental biological characteristics of the animal. The maximum gradient weights correlate exactly to the texture of the dense white fur and the structure of the dog's visage. Significantly, there is no activation in the adjacent grass or vegetation.

This exemplifies a reliable and principled model. This demonstrates that CNN is participating in authentic feature-based learning instead of contextual shortcut learning. Having built and verified our baseline, we can now increase the technological complexity and proceed to the occlusion test to assess the vulnerability of this attention when the major characteristics are removed.

5. Experiment 2: The Occlusion Sensitivity Test

Establishing a healthy baseline in Experiment 1 constitutes merely half of the challenge. To effectively evaluate a deep learning model, we must actively attempt to compromise it. This leads us to the Occlusion Sensitivity Test. By purposefully altering the input data

to obscure the most salient characteristics of the individual, we induce a condition of visual ambiguity in the neural network. The behavior of a model when it is partially "blinded" reveals significantly more about its internal logic than its performance under ideal laboratory settings.

In this experiment, we systematically render a substantial, opaque black bounding box directly over the center of the image. This occlusion entirely obscures the Samoyed's central thorax, chest, and most of its snout—the same geometric regions upon which the model strongly depended during our baseline assessment. Subsequently, we input the compromised image into the ResNet50 architecture to produce a new Grad-CAM heatmap focused on the class "Samoyed."

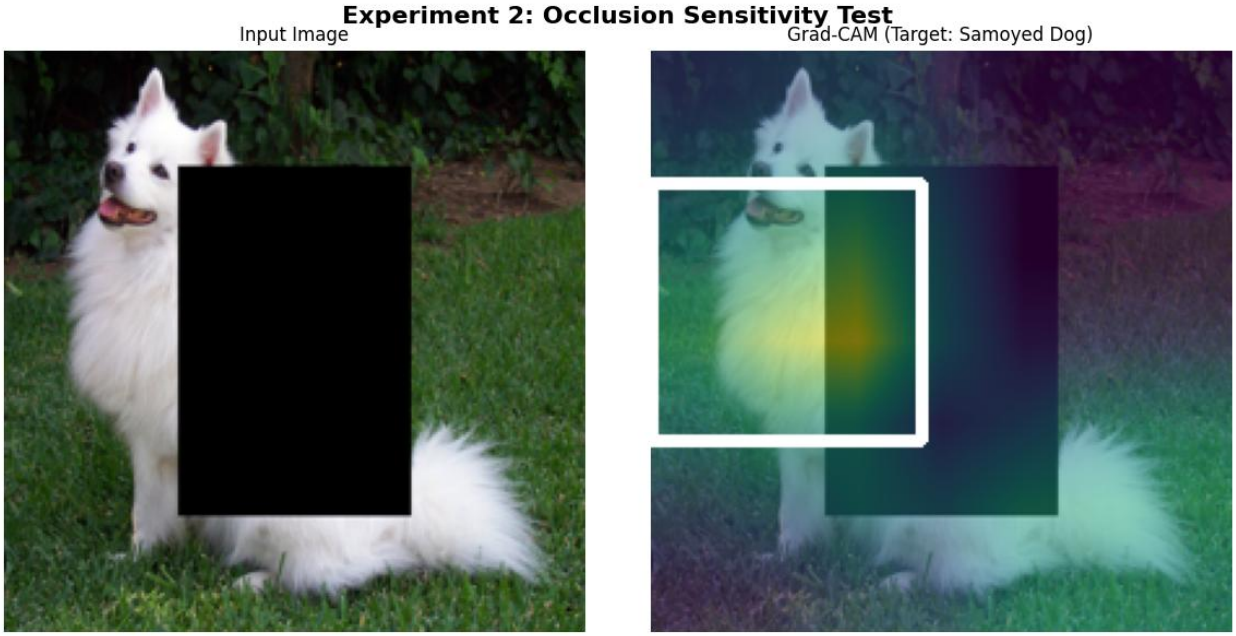

Figure 2 Analysis: Testing Model Resilience
A comparative diagnostic plot. The left panel displays a Samoyed dog, although a substantial black rectangle obscures its midsection, concealing its chest and lower face. The right panel exhibits the Grad-CAM overlay utilizing the Viridis colormap. The substantial white bounding box and the most luminous yellow activations have relocated totally to the left border and top of the black box, prominently emphasizing the remaining sliver of shoulder fur and the dog's left ear.

This outcome exemplifies the "wow" element of CNN interpretability. When stripped of its principal visual references, an inadequately trained or biased model would promptly fail, either predicting an arbitrary class or erratically scanning the background for a Clever Hans shortcut. Our model, however, has exceptional spatial resilience.

The Grad-CAM heatmap identifies the black box as a dead zone. Rather of failing, the network adeptly redirects its gradient attention to the peripheral, focusing on the secondary biological attributes that are still discernible: the distinct texture of the shoulder fur and the triangular shape of the ear.

This demonstrates a crucial mathematical principle: the network has acquired a distributed representation of the target class. It did not merely memorize a singular, central cluster of pixels; it acquired an extensive array of morphological characteristics that delineate a dog. This signifies a highly resilient and reliable model.

What occurs when we eliminate the biological subject and retain solely the contextual background?

6. Experiment 3: Exposing Contextual Hallucinations

In our concluding experiment, we test the network to its limits to reveal its most profound biases. To effectively assess if a model depends on "shortcuts," we must expose it to Out-of-Distribution (OOD) data—images entirely lacking the natural context on which it was trained. We provide the model with a synthetic, abstract geometric figure: a flat purple circle set against a pale green background. This location lacks tangible characteristics.

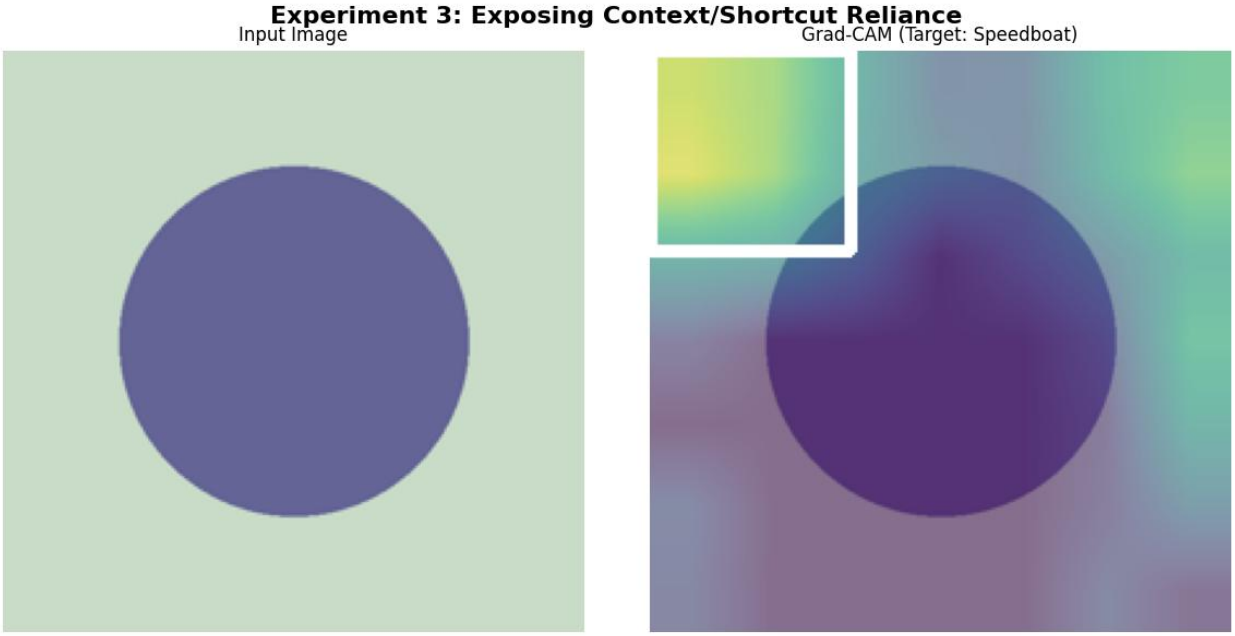

Figure 3 Analysis: The Clever Hans Hallucination
A comparative diagnostic plot. The left panel displays a simple purple circle centered on a light green square. The right panel exhibits the Grad-CAM overlay. The model absolutely forecasted 'Speedboat.' The substantial white bounding box and vivid yellow

activations are wholly segregated in the vacant top-left quadrant of the pale green backdrop, disregarding the purple circle altogether.

This is the precise moment at which the "black box" malfunctions, and it exemplifies the Clever Hans effect that we discussed in the introduction. The model hallucinates the class "Speedboat" when it is compelled to make a prediction based on abstract data.

It is crucial that Grad-CAM demonstrates the precise reason for this: it is examining the empty verdant space. The network has acquired a spurious, perilous correlation. It typically associates the "water" or "sky" that is present in its training images of vessels with flat, light-colored background pixels. It resorted to pure background pattern matching when confronted with visual uncertainty, abandoning the primary subject.

The deployment of a model with this level of contextual bias into high-risk environments, such as autonomous transportation or medical diagnosis, would be catastrophic. Grad-CAM enables us to capture these ethical and technical failures prior to their implementation in production.

7. Conclusion

If a model is dependent on contextual shortcuts rather than genuine feature learning, it is a perilous fallacy to achieve high accuracy. Interpretability tools, such as Grad-CAM and Occlusion Testing, are essential for the auditing of CNNs, as has been demonstrated. By stress-testing the network with missing or out-of-distribution data and visualizing spatial attention, we reveal critical biases prior to deployment. Genuine machine learning proficiency necessitates the demonstration of not only the predictions of a model, but also the rationale behind them.

References

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